

Avengers, Assemble!: We Didn’t Know What You’d Like, So We Put Everything In

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We present the non-standard cosmology suite of the cURAMSES adaptive mesh refinement (AMR) code (cURAMSES III), which assembles, within a single code base, essentially every popular extension of Λ CDM: (i) **self-interacting dark matter** (SIDM) with constant, velocity-dependent, anisotropic and inelastic (two-state) cross-sections, implemented as cell-based Monte Carlo scattering with machine-precision momentum and energy conservation; (ii) a **dark energy and non-GR gravity** battery comprising CPL dark energy with clustering perturbations, massive-neutrino linear response, early dark energy, field-level quintessence and purely kinetic k-essence, coupled quintessence, parametrized quasi-static Horndeski gravity, Hu–Sawicki $f(R)$, nDGP, symmetron, the environmentally damped dilaton, the parameter-free tracker cubic Galileon, and QUMOND — all built on a unified Poisson-solver framework in which model corrections compose multiplicatively, with an FFT-accelerated Newton stage for the nonlinear scalar fields; (iii) **fuzzy dark matter** (FDM), a hybrid Schrödinger–Poisson integrator with a spectral kinetic operator on the base level and finite differences on refined levels; and (iv) **evaporating primordial black holes** (PBH), a table-driven global mass-function scheme with optional local heating. Every dark-energy and gravity module has been verified at the equation level against the literature, at the solver level against analytic linear responses (to 10^{-5} or better), and end-to-end with a controlled 64^3 simulation suite compared against an initial-conditions-consistent linear forward model, agreeing to 2–5 per cent with residual signs explained by screening. We document the algorithms, unit conventions, namelist interfaces and the verification methodology, providing a self-contained reference for non-standard cosmological simulations with AMR codes.

Key words: methods: numerical – cosmology: simulations – dark energy – dark matter – gravitation – neutrinos

1 INTRODUCTION

The Λ CDM concordance model provides an excellent fit to a wide range of cosmological observations (Planck Collaboration 2020), yet the physical nature of both dark components remains unknown. On the dark-energy side, the cosmological constant may be replaced by a dynamical field with an evolving equation of state (Chevallier & Polarski 2001; Linder 2003), by an early dark energy component (Poulin et al. 2019), or the acceleration may signal a breakdown of General Relativity on cosmological scales, as in $f(R)$ gravity (Hu & Sawicki 2007), braneworld models (Schmidt 2009), Galileon (Barreira et al. 2013), symmetron and dilaton theories (Hinterbichler & Khoury 2010; Brax et al. 2010), or MOND (Milgrom 1983). On the dark-matter side, small-scale tensions motivate self-interacting dark matter (Spergel & Steinhardt 2000; Tulin & Yu 2018), ultralight (fuzzy) dark matter (Hui et al. 2017; Schive et al. 2014), and primordial black holes (Carr & Kühnel 2021), while neutrino oscillations guarantee a small but non-zero hot component (Lesgourgues & Pastor 2006).

Stage-IV surveys will constrain all of these at the per-cent level, and their nonlinear signatures can only be predicted by simulations. Historically each extension has lived in its own specialised code; comparing models then convolves physical differences with code systematics. This paper describes the opposite approach: one AMR code, cURAMSES (Kim 2026), carrying *all* of the above extensions behind uniform namelist switches, with shared infrastructure (domain decomposition, Morton-key octree, multigrid/CG/FFT Poisson solvers, OpenMP+MPI parallelism) and a common verification methodology. Hence the title¹.

The paper is organised in four physics sections, one per family: Section 2 describes self-interacting dark matter; Section 3 the dark energy and non-GR gravity battery, including the unified Poisson framework, the nonlinear scalar-field solvers and their verification; Section 4 fuzzy dark matter; and Section 5 evaporating primordial black holes. Section 6 concludes. Throughout, code quantities use the supercomoving variables of Martel & Shapiro (1998): $\tilde{x} = x_{\text{com}}/(aB)$

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¹ From the Korean “*mwol joahalji mollaseo da neoesseo*” — we didn’t know what you’d like, so we put everything in.

(with B the box size), $\tilde{\rho} = \rho a^3/(\rho_c \Omega_m)$ (so $\tilde{\rho} = 1$), $\tilde{\phi} = a^2 \phi/(BH_0)^2$ and $dt = H_0 dt/a^2$.

2 SELF-INTERACTING DARK MATTER

2.1 Physical motivation

SIDM models introduce a non-zero scattering cross-section σ/m that allows dark matter particles to exchange momentum and energy, potentially resolving the core-cusp, missing satellites and too-big-to-fail problems (Spergel & Steinhardt 2000; Tulin & Yu 2018). Astrophysical constraints span dwarf galaxies ($v \sim 30\text{--}100 \text{ km s}^{-1}$) to clusters ($v \sim 10^3\text{--}3 \times 10^3 \text{ km s}^{-1}$); a constant $\sigma/m \sim 1 \text{ cm}^2 \text{ g}^{-1}$ fits dwarf scales but may violate cluster bounds, motivating velocity-dependent cross-sections from light-mediator models (Feng et al. 2009; Tulin et al. 2013; Kaplinghat et al. 2016).

2.2 Monte Carlo scattering algorithm

We implement cell-based Monte Carlo scattering following the standard mesh/SPH approach (Rocha et al. 2013; Robertson et al. 2017). In every AMR leaf cell, dark matter particles are collected from the particle linked list, shuffled with the Fisher–Yates algorithm, and paired sequentially. For a pair with relative velocity v_{rel} the scattering probability per step is

$$P = \frac{\sigma/m(v) \bar{m} v_{\text{rel}} \Delta t}{V_{\text{cell}}} (N_{\text{dm}} - 1), \quad (1)$$

with $\bar{m}$ the mean particle mass, V_{cell} the physical cell volume and the factor $(N_{\text{dm}} - 1)$ accounting for all possible partners. A scatter occurs when a uniform deviate falls below P ; post-scattering velocities are constructed in the centre-of-mass frame,

$$\mathbf{v}'_i = \mathbf{v}_{\text{cm}} + \frac{m_j}{m_i + m_j} |v'_{\text{rel}}| \hat{\mathbf{n}}, \quad \mathbf{v}'_j = \mathbf{v}_{\text{cm}} - \frac{m_i}{m_i + m_j} |v'_{\text{rel}}| \hat{\mathbf{n}}, \quad (2)$$

which conserves momentum by construction and kinetic energy exactly for elastic scattering.

2.3 Cross-sections and angular distributions

Three velocity dependences are supported (`sidm.type`): constant $\sigma/m = \sigma_0$; Yukawa (Born-regime) $\sigma(v)/m = \sigma_0[1 + (v/v_0)^2]^{-2}$; and a generic power law $\sigma(v)/m = \sigma_0(v/v_0)^n$. The angular distribution is isotropic by default, or forward-peaked Rutherford-like, $d\sigma/d\Omega \propto (1 - \cos\theta + 2\varepsilon)^{-2}$ (Kahlhoefer et al. 2014), sampled by CDF inversion.

2.4 Inelastic (two-state) SIDM

A two-state model (Tucker-Smith & Weiner 2001; Vogelsberger et al. 2019) adds a mass splitting δ : ground–ground pairs can up-scatter ($g+g \rightarrow e+e$, allowed when $\frac{1}{2}\mu v_{\text{rel}}^2 > \delta$), excited pairs down-scatter exothermically, and mixed pairs scatter elastically; the relative speed is adjusted to conserve total (kinetic + internal) energy. An initial excited fraction is assigned at start-up in a restart-safe manner. A dissipative channel (`sidm.fdis`) and a DM–baryon drag term (`sidm.baryon`) are also available.

2.5 Timestep, threading and diagnostics

The maximum per-level probability P_{max} is fed back into the timestep as $\Delta t_{\text{new}} \leq (C_{\text{SIDM}}/P_{\text{max}}) \Delta t_{\text{old}}$ (default $C_{\text{SIDM}} = 0.1$). The sweep is OpenMP-parallel over grids with per-thread decorrelated RNG streams and deterministic reseeding, and cumulative momentum/energy residuals are reduced across MPI ranks each step, providing runtime verification of the conservation properties. All parameters live in `&SIDM_PARAMS`.

3 DARK ENERGY AND NON-GR GRAVITY

This section describes the largest part of the suite: eleven dark-energy and modified-gravity models sharing one solver infrastructure. Table 1 summarises the models, their parameters and their verification status.

3.1 Unified Poisson-solver framework

In supercomoving units the code solves $\tilde{\nabla}^2 \tilde{\phi} = \frac{3}{2} \Omega_m a (\tilde{\rho} - 1)$. All smooth-dark-sector corrections enter this equation as multiplicative factors on the source. A single function, `cosmo_poisson_fourpi()`, provides the prefactor consistently to the multigrid, conjugate-gradient and coarse-level solvers, and the FFT solvers apply the same corrections per Fourier mode in the Green’s function, so that scale-dependent responses cost nothing extra. The factors compose:

$$\frac{3}{2} \Omega_{cb} a \times \underbrace{\left[1 + \frac{\Omega_\nu}{\Omega_{cb}} R_\nu(k, a)\right]}_{\text{neutrinos}} \underbrace{\left[1 + \frac{\Omega_{\text{DE}}(a)}{\Omega_{cb}} R_{\text{DE}}(k, a)\right]}_{\text{DE clustering}} \underbrace{\mu(a, k)}_{\text{Horndeski coupled DE}} \underbrace{e^{-\beta(\varphi - \varphi_0)}}_{\text{DE}}. \quad (3)$$

3.2 CPL dark energy with clustering perturbations

The background uses the exact CPL integral $f_{\text{DE}}(a) = a^{-3(1+w_0+w_a)} e^{-3w_a(1-a)}$ in the supercomoving Friedmann factors, the growth factor and the initial velocity scaling. Perturbations are supported in two modes:

(i) *Table mode*. A CAMB-derived table of the effective ratio $R_{\text{DE}}(k, a) = \delta_{\text{DE}}/\delta_m$ is measured with the Weyl-potential method (the ratio of potential per unit matter contrast between a clustering-DE and a smooth-DE universe), which yields the Newtonian-gauge response appropriate for the Poisson source; synchronous-gauge δ_{DE} would over-correct by an order of magnitude.

(ii) *Quasi-static fallback*. When no table is supplied and $c_s^2 > 0$, the code applies the Sapone & Kunz (2009) closure

$$\frac{\delta_{\text{DE}}}{\delta_m} = \frac{1 + w}{(1 - 3w) + \frac{2c_s^2 k^2}{3a^2 H^2}}, \quad \text{i.e. d.f.} = 1 + \frac{\alpha}{k^2 + \kappa^2}, \quad (4)$$

with $\kappa^2 = \frac{3}{2}(1 - 3w)a^2 E^2(L/\lambda_H)^2/c_s^2$ and $\alpha = (\Omega_\Lambda f_{\text{DE}} a^3/\Omega_m)(1 + w) \frac{3}{2} a^2 E^2(L/\lambda_H)^2/c_s^2$. Correct limits: a k -independent boost $1 + (\rho_{\text{DE}}/\rho_m)(1 + w)/(1 - 3w)$ for $c_s^2 \rightarrow 0$, unity below the sound horizon, and bit-wise unity for Λ . Validated against CAMB (same Weyl method) to ≤ 0.1 per cent for $(w_0, w_a) = (-0.9, 0.1)$ and ≤ 0.8 per cent for $w_0 = -0.8$ at $z \geq 0.5$ (~ 4 per cent at $z = 0$, where the quasi-static approximation degrades — use the table mode for precision).

Table 1. The dark energy and gravity battery. “Linear response” lists the analytically validated unscreened limit; the last column the accuracy achieved in dedicated unit tests of the discretised equations (see Section 3.10).

Model	Switch / namelist	Background	Linear response	Unit-test accuracy
CPL DE + perturbations	w0,wa,cs2_de,de_table	$w = w_0 + w_a(1 - a)$	Sapone–Kunz closure	$\leq 0.1\text{--}0.8\%$ vs CAMB ($z \geq 0.5$)
Massive ν linear response	use_neutrino	—	$1 + (\Omega_\nu/\Omega_{cb})R_\nu(k, a)$	table-exact
Early dark energy	use_ede	Poulin+19 fluid	smooth	analytic dilution exact
Quintessence	use_quintessence	Klein–Gordon (shooting)	smooth ($c_s^2 = 1$)	w -consistency 2×10^{-5}
k-essence	use_kessence	$P(X) = -X + X^2$ (exact)	Helmholtz with $c_s^2(a)$	analytic limits exact
Coupled quintessence	use_coupled_de	KG + mass evolution	$G_{\text{eff}} = G(1 + 2\beta^2)$	isolation runs $\leq 1\%$
Horndeski $\mu(a, k)$	use_horndeski	Λ CDM/CPL	$\mu = 1 + \mu_0\Omega_{\text{DE}}(a)/\Omega_{\text{DE},0}$	exact by construction
$f(R)$ Hu–Sawicki	use_fR	Λ CDM	$\mu = 1 + \frac{1}{3} \frac{k^2}{k^2 + m^2 a^2}$	6×10^{-7}
nDGP	use_nDGP	Λ CDM	$G_{\text{eff}}/G = 1 + 1/(3\beta)$	force field 1.5×10^{-6}
Symmetron	use_symmetron	Λ CDM	$F_5/F_N = 2\beta^2\chi^2$	exact (Brax+12 eq. 68 match)
Dilaton (Brax+12)	use_dilaton	Λ CDM	$2\beta(a)^2 k^2/(k^2 + m^2 a^2)$	5 digits
Cubic Galileon (tracker)	use_galileon	tracker $E^2(a)$	$G_{\text{eff}}/G(a=1) = 1.844$	5 digits
QUMOND	use_mond	—	$\nu(g_N /a_0)$	$\mu\nu$ conjugacy 2×10^{-16}

at low redshift). An earlier release used a closure with the opposite sign (a 15–29 per cent source suppression); results obtained with it should be discarded.

3.3 Massive neutrino linear response

Neutrino perturbations are added without neutrino particles: $\delta_\nu(k, a) = R_\nu(k, a) \delta_{cb}(k, a)$ with R_ν precomputed from CAMB (Lewis et al. 2000), entering the Poisson source as in Section 3.1. This avoids neutrino shot noise entirely and is exact at linear order (cf. Ali-Haïmoud & Bird 2013).

3.4 Early dark energy

The Poulin et al. (2019) fluid form $\rho_{\text{EDE}}(a) = \omega_{\text{ede}} \rho_{\text{std}}(a_c) 2/[1 + (a/a_c)^{3(1+w_{\text{ede}})}]$ is added to the Friedmann factors. The implementation reproduces the analytic dilution slope $-3(1+w)$ and the analytic peak $f_{\text{EDE}}^{\text{max}} = 1.14\omega_{\text{ede}}$ at $a = 3^{1/4}a_c$ exactly. Two conventions matter when mapping CMB constraints: ω_{ede} is the fraction at a_c (not the peak), and the background carries no radiation, so at $z_c \sim 3000$ the fraction differs from Planck-style f_{EDE} by $\sim 2\times$.

3.5 Scalar-field dark energy: quintessence and k-essence

A dedicated module tabulates scalar-field backgrounds on a $\ln a$ grid ($a = 10^{-6}\text{--}1.55$) with lazy, self-healing initialisation (the tables rebuild automatically if the cosmology is overridden by IC headers).

Quintessence (Ratra & Peebles 1988): the Klein–Gordon and Friedmann equations are integrated with RK4 for Ratra–Peebles ($V \propto \varphi^{-\alpha}$) or exponential potentials, the amplitude fixed by shooting to $\Omega_\varphi(a=1) = \Omega_\Lambda$. The tabulated $w(a)$ obeys $d \ln f_{\text{DE}}/d \ln a = -3(1+w)$ to 2×10^{-5} , and the RP tracker attractor $w = -2/(\alpha + 2)$ is recovered in the matter era.

k-essence (Armendariz-Picon et al. 2001): for the purely kinetic $P(X) = -X + X^2$ the shift-symmetry integral $P_{,X}\sqrt{2X}a^3 = \text{const}$ gives $X(a)$ algebraically; $w \rightarrow 1/3$ and $c_s^2 \rightarrow 1/3$ at early times (a dark-radiation-like component) and $w \rightarrow -1$ today. The model’s $c_s^2(a)$ feeds the Helmholtz closure of equation (4) automatically.

3.6 Coupled quintessence

Conformally coupled dark matter, $m_{\text{dm}} \propto e^{-\beta\varphi/M_{\text{Pl}}}$ (Amen-dola 2000), adds four effects, all implemented and individually verified: (i) the modified expansion, including the DM mass-evolution factor in the *matter* term of the Friedmann integrals (its omission is detectable: an isolation run reproduced the inconsistent- H growth prediction to 0.2 per cent); (ii) the same factor multiplying the Poisson source; (iii) the momentum-conservation velocity term $+\beta\dot{\varphi}\mathbf{v}$, split across the two half-kicks of the KDK integrator; and (iv) the fifth-force enhancement $G_{\text{eff}} = G(1 + 2\beta^2)$ between matter particles.

3.7 Parametrized Horndeski gravity

The quasi-static, scale-independent limit of Horndeski gravity is implemented as $\mu(a) = 1 + \mu_0\Omega_{\text{DE}}(a)/\Omega_{\text{DE},0}$ multiplying the Poisson source, with an optional Compton-mass k -dependence $\mu(a, k) = 1 + [\mu(a) - 1]k^2/(k^2 + a^2m^2)$ applied exactly in the FFT solver paths (the multigrid path uses the $k \rightarrow \infty$ limit).

3.8 Screened modified gravity: the scalar-field solvers

Five models solve a nonlinear elliptic equation for an extra scalar degree of freedom on the AMR hierarchy, with the fifth force added to the particle acceleration.

3.8.1 $f(R)$ Hu–Sawicki

The quasi-static equation in code units is

$$\tilde{\nabla}^2 f_R = \frac{a^2}{3} \left(\frac{H_0 B}{c} \right)^2 [\tilde{R}(f_R) - \tilde{R}] - \Omega_m \left(\frac{H_0 B}{c} \right)^2 \frac{\tilde{\rho} - 1}{a}, \quad (5)$$

with the Hu–Sawicki inversion $\tilde{R} = \tilde{R}_0(f_{R0}/f_R)^{1/(n+1)}$ and force $F_5 = +\frac{a^2}{2}(c/H_0 B)^2 \tilde{\nabla}^2 f_R$ (Hu & Sawicki 2007; Oyaizu 2008; Li et al. 2012). The linearized mass term is a Yukawa (the Jacobian is negative definite) and the chameleon limit $\tilde{R} \rightarrow \tilde{R} + 3\Omega_m \delta/a^3$ is exact. The discretised linear response reproduces $\mu(k) - 1 = \frac{1}{3}k^2/(k^2 + m^2 a^2)$ to 6×10^{-7} across $a \in [0.25, 1]$, $f_{R0} \in \{-10^{-5}, -10^{-6}\}$, $n \in \{1, 2\}$.

3.8.2 nDGP

The brane-bending mode obeys the Vainshtein equation (Schmidt 2009; Li et al. 2013); in code units

$$\tilde{\nabla}^2 \varphi + \frac{1}{12 \Omega_{rc} \beta(a) a^4} [(\tilde{\nabla}^2 \varphi)^2 - (\tilde{\nabla}_i \tilde{\nabla}_j \varphi)^2] = \frac{\Omega_m a}{\beta(a)} (\tilde{\rho} - 1), \quad (6)$$

with $\beta = 1 + 2Hr_c(1 + \dot{H}/3H^2)$ and $F_5 = -\frac{1}{2}\tilde{\nabla}\varphi$, so that the unscreened limit is $G_{\text{eff}}/G = 1 + 1/(3\beta)$ (Winther et al. 2015).

3.8.3 Symmetron

The dimensionless field $\chi = \varphi/\varphi_*$ obeys $\tilde{\nabla}^2 \chi = \frac{a^2}{2\tilde{\chi}^2} [\tilde{\rho}(a_{\text{ssb}}/a)^3 - 1]\chi + \frac{a^2}{2\tilde{\chi}^2} \chi^3$ (Hinterbichler & Khourey 2010; Davis et al. 2012), identical to eq. (68) of Brax et al. (2012), with force $F_5 = -6\Omega_m \beta^2 (\lambda/B)^2 (a^2/a_{\text{ssb}}^3) \chi \tilde{\nabla}\chi$, whose unscreened limit is the canonical $F_5/F_N = 2\beta^2 \tilde{\chi}^2$. Because $\chi = 0$ is an exact fixed point of the homogeneous equation, the broken-phase VEV $\bar{\chi}(a) = \sqrt{1 - (a_{\text{ssb}}/a)^3}$ is seeded into uninitialised cells at every solve (this also covers restarts and newly refined cells).

3.8.4 Environmentally damped dilaton

The original $r = 3/2$ dilaton of Brax et al. (2010), with $A(\varphi) = 1 + \frac{A_2}{2}\varphi^2/M_{\text{Pl}}^2$, $m^2(a) = 3A_2 H^2(a)$ and $\beta(a) = \beta_0 a^{3\Omega_m}$ (Brax et al. 2012), is solved in $\chi = \varphi/M_{\text{Pl}}$:

$$\tilde{\nabla}^2 \chi = \frac{3\Omega_m A_2 B_2}{a} (\tilde{\rho} \chi - \bar{\chi}) + a^2 B_2 [v_\varphi(\chi) - v_\varphi(\bar{\chi})], \quad (7)$$

with $v_\varphi(\chi) = -3\Omega_m \beta_0 (A_2 \chi / \beta_0)^{1-3/s}$, $s = 3\Omega_m$, $B_2 = (H_0 B/c)^2$ and $F_5 = -(c/H_0 B)^2 a^2 A_2 \chi \tilde{\nabla}\chi$. The exponent $1 - 3/s < 0$ makes the Newton Jacobian negative definite. The background is an exact fixed point of the discrete operator, the linear response matches $2\beta(a)^2 k^2/(k^2 + m^2 a^2)$ to five digits, and a $\delta = 10^5$ top-hat drives $\chi \rightarrow 0.03\bar{\chi}$ (Damour–Polyakov screening). User parameters are simply β_0 and the present force range $\lambda = 2998 \xi h^{-1} \text{Mpc}$ (then $A_2 = 1/3\xi^2$).

3.8.5 Tracker cubic Galileon

On the tracker, $H\dot{\varphi} = \xi H_0^2 M_{\text{Pl}}$ with $c_2 = -6c_3\xi$, the model is parameter-free: $\xi = \sqrt{6(1 - \Omega_m)}$, $E^2(a) = \frac{1}{2}[\Omega_m a^{-3} + \sqrt{\Omega_m^2 a^{-6} + 4(1 - \Omega_m)}]$ (wired through the background dispatch, so the modified expansion and growth are automatic; $w_0 \simeq -1.18$), and the perturbation coefficients of Barreira et al. (2013) reduce to $\beta_1 = \frac{\xi}{3}[2\dot{H}/H^2 - 1 + (1 - \Omega_m)/E^4]$, $\beta_2 = 2E^2\beta_1/\xi^2$. The field equation is the nDGP template with coefficient $1/(3\beta_1 a^4)$ and source $\Omega_m a \delta/\beta_2$; the Poisson back-reaction term integrates to a potential proportional to the field, so the entire fifth force is one gradient, $F_5 = +\frac{\xi}{6E^2}\tilde{\nabla}u$. The unscreened response matches the analytic $-\xi/(9\beta_2 E^2)$ to five digits at $a = 0.25$ –1: $G_{\text{eff}}/G(a=1) = 1.844$, decaying as $1/E^4$ into the past. Since $G_{\text{eff}}/G - 1 < 10^{-3}$ for $z \gtrsim 2.5$, the solver is skipped at early times.

3.8.6 QUMOND

The quasi-linear formulation of MOND (Milgrom 2010) defines a curl-free field via $\nabla^2 \Phi = \nabla \cdot [\nu(|\nabla \Phi_N|/a_0) \nabla \Phi_N]$. Both

an algebraic force-correction mode and the full phantom-density mode (compute $\rho_{\text{ph}} = \nabla \cdot [(\nu - 1)\nabla \Phi_N]/4\pi G$, re-solve Poisson) are implemented, with the simple and standard interpolation pairs (verified mutually conjugate to 2×10^{-16}), an external-field option, and a_0 converted to code units through the exact supercomoving acceleration factor (so the MOND scale is correctly constant in proper units at all redshifts). An AQUAL fixed-point mode is also available (cf. Lügghausen et al. 2015; Candlish et al. 2015).

3.9 Numerical solution of the scalar equations

All five scalar solvers share one machinery: nonlinear Newton–Gauss–Seidel relaxation with true three-dimensional red–black colouring (cell parity $i+j+k$), Neumann closure of the stencil at coarse–fine boundaries, one-sided force gradients at level edges, and MPI ghost exchange each sweep. On the fully refined domain level the long-wavelength modes — which plain relaxation cannot converge in $\mathcal{O}(10)$ sweeps — are solved spectrally: a Newton–Helmholtz stage ($\tilde{\nabla}^2 - \bar{m}^2$) $\delta u = -\mathcal{R}$ with the level-averaged Jacobian mass for $f(R)$ /symmetron/dilaton, and a Winther et al. (2015)-style operator splitting for the Vainshtein-type models (solve the cell-local quadratic for $\tilde{\nabla}^2 \varphi$, then invert the Laplacian by FFT). In a controlled test the operator-split stage reached a lower residual in five iterations than 2×10^4 damped relaxation sweeps. Two MPI subtleties deserve mention as a warning to implementers: every rank must enter the per-level ALLREDUCE of the convergence test and the final force ghost-exchange *even when it owns no grids on that level*; early returns on empty ranks deadlock or abort the run at the first sparsely refined level.

3.10 Verification

Verification proceeded on three levels.

Equation level. Every model was re-derived from its original reference (unit conversions included) and the code compared term by term; this stage found and fixed, among others, a wrong-sign DE-perturbation closure, a tachyonic (non-convergent) $f(R)$ operator, a double-counted nDGP coupling, a symmetron force normalisation missing the box-unit conversion, and a zero-pinned symmetron field.

Solver level. The discretised operators were exercised in isolation against analytic linear responses; the accuracies are listed in Table 1.

End-to-end level. A controlled suite of 64^3 , $100 h^{-1} \text{Mpc}$ dark-matter-only simulations was run from shared Zel’dovich initial conditions ($z = 49$, CAMB spectrum) for ΛCDM , quintessence, k-essence, Horndeski, coupled DE (plus per-effect isolation runs), the tracker Galileon and the dilaton. The measured $P_{\text{model}}/P_{\Lambda\text{CDM}}$ in the band $k = 0.05$ – $0.12 h \text{Mpc}^{-1}$ at $z \simeq 0$ was compared against an *initial-conditions-consistent* linear forward model, which (i) rescales the initial displacements by each model’s own velocity conversion factor — RAMSES converts the shared velocity files with the model’s v_{fact} , an effect reaching $(1.077)^2$ for k-essence and $(1.109)^2$ for coupled DE — and (ii) propagates the exact (δ, θ) initial vector through each model’s linear system. Results:

All models agree to 2–5 per cent; the Galileon and dilaton deficits carry the expected sign, since Vainshtein and

Table 2. End-to-end suite: simulated vs. predicted $P/P_{\Lambda\text{CDM}}$ ($k = 0.05\text{--}0.12\ h/\text{Mpc}$, $z \simeq 0$).

Model	Simulation	Forward model	Ratio
Quintessence (RP $\alpha=1$)	0.832	0.817	1.018
k-essence ($X_0/M^4=0.5005$)	0.875	0.881	0.993
Horndeski ($\mu_0=0.2$)	1.036	1.052	0.985
Coupled DE ($\beta=0.1$)	0.966	1.005	0.961
Galileon (tracker)	1.148	1.208	0.950
Dilaton ($\beta_0=0.5$, $\lambda=20$)	1.039	1.071	0.970

Damour–Polyakov screening remove part of the fifth-force boost already at quasi-linear wavenumbers (cf. Barreira et al. 2013). Each coupled-DE ingredient (mass evolution, friction, G_{eff}) was additionally isolated with toggle runs and verified at the ~ 1 per cent level. The forward model, the IC generator and all unit-test drivers ship with the code (`patch/cuRamses/aux/`).

4 FUZZY DARK MATTER

Ultralight scalar dark matter with $m_a \sim 10^{-22}\text{ eV}$ (Hui et al. 2017) is evolved as a classical field ψ obeying the Schrödinger–Poisson system. The integrator uses drift–kick–drift operator splitting per AMR level: the kinetic half-steps apply $\exp(-i\hbar k^2 \Delta t / 4a^2 m_a)$ spectrally on the base level (FFT) and an explicit finite-difference Laplacian on refined levels, while the potential kick applies $\exp(-im_a \Phi \Delta t / \hbar)$ in real space; a fourth-order Yoshida composition is optional (`fdm_split_order=4`). Refinement follows the de Broglie wavelength (a minimum of `fdm_nrefine_dB` cells per λ_{dB} , with a density floor), reproducing the solitonic cores and interference fringes characteristic of FDM haloes (Schive et al. 2014). A hybrid mode (`fdm_use_hjm`) evolves a Hamilton–Jacobi–Madelung fluid on coarse levels and switches to the wave solver above `fdm_first_wave_level`, and `fdm_hybrid` allows mixed FDM + particle (stars/sinks) runs. The wavefunction is stored as $(\psi_{\text{re}}, \psi_{\text{im}})$ on the AMR mesh with $|\psi|^2$ the total DM density, and dedicated light-cone and output modules are provided. A detailed description and validation of the FDM solver is presented in a companion paper.

5 EVAPORATING PRIMORDIAL BLACK HOLES

The PBH module models dark matter that is partly composed of Hawking-evaporating primordial black holes (Hawking 1975; Carr & Kühnel 2021). Rather than tracking per-particle masses, it uses one global mass function: all DM particles share the mixed-mass factor

$$w(a) = 1 - f_{\text{PBH}} + f_{\text{PBH}} g(a), \quad (8)$$

where $g(a)$ is the surviving PBH mass fraction from a pre-computed evaporation table and f_{PBH} (`pbh_fraction`) the initial PBH share of the dark matter. Particle masses are rescaled exactly by table arithmetic — there is no per-particle state and no evaporation CFL constraint. The energy released between steps follows the cumulative heating kernel $\tilde{Q}(a) = \int g g dt$; it is either deposited as local heat in the gas (`pbh_energy_sink='local_heat'`, optionally amplified by a

linear-response boost factor) or removed from the simulation (`'removed'`, appropriate for free-streaming decay products). Restart provenance (table checksum, normalisation epoch) is persisted so that a run cannot silently continue with a different evaporation model. The table generator supports arbitrary monochromatic or extended PBH mass functions.

6 CONCLUSIONS

We have assembled, in one AMR code, the four families of non-standard cosmological physics most often invoked beyond ΛCDM — self-interacting dark matter, a comprehensive dark-energy and modified-gravity battery, fuzzy dark matter, and evaporating primordial black holes — behind uniform namelist switches and on shared solver infrastructure. Beyond the implementations themselves, we highlight the verification methodology: every model is checked at the equation level against its original reference, at the solver level against analytic linear responses, and end-to-end against an initial-conditions-consistent forward model, with per-effect isolation runs where a model bundles several couplings. We commend this three-level protocol to implementers of non-standard physics in any simulation code: in our experience each level catches defects invisible to the others, ranging from wrong-sign perturbation closures to factor-of-two integrator bookkeeping and MPI collective mismatches that only manifest on sparsely refined levels.

All modules recover ΛCDM bit-wise when disabled. The namelist interfaces, verification scripts and the simulation-suite tooling are distributed with the code.

ACKNOWLEDGEMENTS

The author thanks the KIAS Center for Advanced Computation for computing resources. Portions of the implementation and the verification campaign were carried out with the assistance of Claude (Anthropic).

DATA AVAILABILITY

The code, namelist examples, verification drivers and the 64^3 test-suite configuration are available from the `CURAMSES` repository. Simulation outputs will be shared on reasonable request.

REFERENCES

- Ali-Haïmoud Y., Bird S., 2013, *MNRAS*, 428, 3375
- Amendola L., 2000, *Phys. Rev. D*, 62, 043511
- Armendariz-Picon C., Mukhanov V., Steinhardt P. J., 2001, *Phys. Rev. D*, 63, 103510
- Barreira A., Li B., Hellwing W. A., Baugh C. M., Pascoli S., 2013, *J. Cosmology Astropart. Phys.*, 10, 027
- Brax P., van de Bruck C., Davis A.-C., Shaw D., 2010, *Phys. Rev. D*, 82, 063519
- Brax P., Davis A.-C., Li B., Winther H. A., Zhao G.-B., 2012, *J. Cosmology Astropart. Phys.*, 10, 002
- Candlish G. N., Smith R., Fellhauer M., 2015, *MNRAS*, 446, 1060
- Carr B., Kühnel F., 2021, *arXiv:2110.02821*
- Chevallier M., Polarski D., 2001, *Int. J. Mod. Phys. D*, 10, 213

- Davis A.-C., Li B., Mota D. F., Winther H. A., 2012, *ApJ*, 748, 61
- Feng J. L., Kaplinghat M., Tu H., Yu H.-B., 2009, *J. Cosmology Astropart. Phys.*, 07, 004
- Hawking S. W., 1975, *Comm. Math. Phys.*, 43, 199
- Hinterbichler K., Khoury J., 2010, *Phys. Rev. Lett.*, 104, 231301
- Hu W., Sawicki I., 2007, *Phys. Rev. D*, 76, 064004
- Hui L., Ostriker J. P., Tremaine S., Witten E., 2017, *Phys. Rev. D*, 95, 043541
- Kahlhoefer F., Schmidt-Hoberg K., Frandsen M. T., Sarkar S., 2014, *MNRAS*, 437, 2865
- Kaplinghat M., Tulin S., Yu H.-B., 2016, *Phys. Rev. Lett.*, 116, 041302
- Kim J., 2026, *MNRAS*, submitted (cuRAMSES I)
- Lesgourgues J., Pastor S., 2006, *Phys. Rep.*, 429, 307
- Lewis A., Challinor A., Lasenby A., 2000, *ApJ*, 538, 473
- Li B., Zhao G.-B., Teyssier R., Koyama K., 2012, *J. Cosmology Astropart. Phys.*, 01, 051
- Li B., Zhao G.-B., Koyama K., 2013, *J. Cosmology Astropart. Phys.*, 05, 023
- Linder E. V., 2003, *Phys. Rev. Lett.*, 90, 091301
- Llinares C., Mota D. F., Winther H. A., 2014, *A&A*, 562, A78
- Lüghausen F., Famaey B., Kroupa P., 2015, *Canadian J. Phys.*, 93, 232
- Martel H., Shapiro P. R., 1998, *MNRAS*, 297, 467
- Milgrom M., 1983, *ApJ*, 270, 365
- Milgrom M., 2010, *MNRAS*, 403, 886
- Oyaizu H., 2008, *Phys. Rev. D*, 78, 123523
- Planck Collaboration, 2020, *A&A*, 641, A6
- Poulin V., Smith T. L., Karwal T., Kamionkowski M., 2019, *Phys. Rev. Lett.*, 122, 221301
- Ratra B., Peebles P. J. E., 1988, *Phys. Rev. D*, 37, 3406
- Robertson A., Massey R., Eke V., 2017, *MNRAS*, 465, 569
- Rocha M., Peter A. H. G., Bullock J. S., et al., 2013, *MNRAS*, 430, 81
- Sapone D., Kunz M., 2009, *Phys. Rev. D*, 80, 083519
- Schive H.-Y., Chiueh T., Broadhurst T., 2014, *Nature Physics*, 10, 496
- Schmidt F., 2009, *Phys. Rev. D*, 80, 043001
- Spergel D. N., Steinhardt P. J., 2000, *Phys. Rev. Lett.*, 84, 3760
- Teyssier R., 2002, *A&A*, 385, 337
- Tucker-Smith D., Weiner N., 2001, *Phys. Rev. D*, 64, 043502
- Tulin S., Yu H.-B., 2018, *Phys. Rep.*, 730, 1
- Tulin S., Yu H.-B., Zurek K. M., 2013, *Phys. Rev. D*, 87, 115007
- Vogelsberger M., Zavala J., Schutz K., Slatyer T. R., 2019, *MNRAS*, 484, 5437
- Winther H. A., et al., 2015, *MNRAS*, 454, 4208